ملخص الفصل 5: خمس أنواع أسئلة

في السؤال	النوع	أول سطر
If . . ., then . . . أو For every . . . + Disprove	1. مثال مضاد	For $n = \ldots, \ldots$
irrational أو There is no . . . + Prove	2. تناقض	Assume, to the contrary, . . .
by (a) direct, (b) contrapositive, (c) . . .	3. ثلاث طرق	جدول النوع 3
There exists . . . + Prove	4. وجود	Let $x = \ldots$ / Consider . . .
There exists . . . + Disprove	5. نفي وجود	We show that if . . ., then . . .
Prove or disprove	جرّب 3 قيم أول، وبعدها قرّر النوع	

1 المثال المضاد

تعرفه: For every + Disprove

جرّب ← احسب ← احكم

مثال For every even positive integer n, $12 \mid (n^2 - 4)$.

جرّب: $n = 2$ يعطي 0 ✓ · $n = 4$ يعطي 12 ✓ · $n = 6$ يعطي 32 ✗

الجواب **False.** For $n = 6$, $n^2 - 4 = 32$ and $12 \nmid 32$. Thus, $n = 6$ is a counterexample.

⚠️ أول قيمة أو قيمتين ممكن يمشون: جرّب 3 قيم. وفي «إذا... فإن...» المثال المضاد: الشرط صح والنتيجة خطأ.

2 البرهان بالتناقض

تعرفه: irrational أو There is no

نعكس ← نمشي ← تناقض

مثال 1 There is no smallest positive real number.

الجواب Assume, to the contrary, that there is a smallest positive real number, say r. Since $0 < r/2 < r$, $r/2$ is a positive real number smaller than r. This is a contradiction.

مثال 2 Prove that $\sqrt{2}$ is irrational.

الجواب Assume, to the contrary, that $\sqrt{2}$ is rational. Then $\sqrt{2} = a/b$, where $a, b \in \mathbb{Z}$, $b \neq 0$, and a/b is in lowest terms. So $a^2 = 2b^2$, and a is even. Let $a = 2c$. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$ and b is even. Since a and b are both even, this contradicts that a/b is in lowest terms.

⚠️ **العكس:** «ما فيه» ← «فيه» · «غير نسبي» ← «نسبي» · «إذا P فإن Q» ← «Q صح و P خطأ» · «و» ← «أو».

5 جمل تحفظها

- Thus, $n = \ldots$ is a counterexample.
- Assume, to the contrary, that . . .
- This is a contradiction.
- . . . has the desired property.
- We show that if . . ., then . . .

3 ثلاث طرق

تعرفه: by (a) direct, (b) contrapositive, (c) contradiction

مباشر	عكس النقيض	تناقض
افترض P ← وصّل لـ Q	افترض عكس Q ← وصّل لعكس P	افترض P وعكس Q ← مستحيل

مثال

If n is an even integer, then $5n + 1$ is odd.

(a) مباشر

Assume that n is even. Then $n = 2k$ for some integer k. So $5n + 1 = 10k + 1 = 2(5k) + 1$, which is odd.

(b) عكس النقيض

Assume that $5n + 1$ is even. Then $5n + 1 = 2k$ for some integer k. So $n = (5n + 1) - 4n - 1 = 2(k - 2n - 1) + 1$, which is odd.

(c) تناقض

Assume, to the contrary, that there is an even integer n such that $5n + 1$ is even. Then $n = 2x$ and $5n + 1 = 2y$ for some integers x, y. So $10x + 1 = 2y$ and $1 = 2(y - 5x)$. Thus 1 is even, which is a contradiction.

⚠️ لا تبدأ أبداً بافتراض Q (أن $5n + 1$ فردي). هذا افتراض للمطلوب.

4 الوجود

تعرفه: There exists + Prove

اختر ← تحقق ← اختم

مثال 1

There exists an integer whose cube equals its square.

الجواب

Since $1^3 = 1^2 = 1$, the integer 1 has the desired property.

مثال 2

The equation $x^5 + 2x - 5 = 0$ has a unique real solution between $x = 1$ and $x = 2$.

الجواب

Existence. Let $f(x) = x^5 + 2x - 5$. Since f is a polynomial, it is continuous on $[1, 2]$. Since $f(1) = -2 < 0 < 31 = f(2)$, by the Intermediate Value Theorem there is $c \in (1, 2)$ with $f(c) = 0$.

Uniqueness. Assume, to the contrary, that there are two solutions $a < b$ in $(1, 2)$. Then $a^5 + 2a - 5 < b^5 + 2b - 5$, so $0 < 0$, a contradiction.

⚠️ «حل وحيد» = جزأين: الوجود بالقيمة الوسطى، والوحدانية بالتناقض.

5 نفي الوجود

تعرفه: There exists + Disprove

اقلب ← خذ أي x ← اثبت

مثال

Disprove: There exists an integer n such that $n^3 - n + 1$ is even.

الجواب

We show that $n^3 - n + 1$ is odd for every $n \in \mathbb{Z}$.

Case 1. $n = 2a$: $n^3 - n + 1 = 8a^3 - 2a + 1 = 2(4a^3 - a) + 1$, which is odd.

Case 2. $n = 2b + 1$: $n^3 - n + 1 = 8b^3 + 12b^2 + 4b + 1 = 2(4b^3 + 6b^2 + 2b) + 1$, which is odd.

⚠️ هنا مثال واحد ما يكفي: لازم تثبت «لكل n».